3AL Tutorial 8 2024 Solutions

1. Let X be the set of subgroups of a group G. Prove that conjugation is a group action on X.

We need to show that if $H \in X$ is a subgroup of G and $g \in G$, then $g * H = gHg^{-1}$ is also a subgroup of G. It is clear that $gHg^{-1} \subseteq G$. Note that $1 = g1g^{-1} \in gHg^{-1}$ so gHg^{-1} is nonempty. Take $a, b \in gHg^{-1}$. So $a = ghg^{-1}$ and $b = gh'g^{-1}$ for some $h, h' \in H$. Then

$$ab^{-1} = (ghg^{-1})(gh'g^{-1})^{-1} = g(hh'^{-1})g^{-1} \in gHg^{-1}.$$

Thus gHg^{-1} is a subgroup of G and so conjugation by G is well-defined on X.

Take $g_1, g_2 \in G$ and $H \in X$. Then

$$g_1 * (g_2 * H) = g_1(g_2 H g_2^{-1})g_1^{-1} = (g_1g_2)H(g_1g_2)^{-1} = g_1g_2 * H$$
$$1 * H = 1H1^{-1} = H.$$

Thus conjugation is a group action on X.

2. Let G be a group with subgroup H. Prove that $H \lhd G$ if and only if H is a union of conjugacy classes.

Suppose $H \lhd G$. For all $a \in H$, we have $a \in \operatorname{class} a$. Thus $H \subseteq \bigcup_{a \in H} \operatorname{class} a$. We need to show that $\bigcup_{a \in H} \operatorname{class} a \subseteq H$. Let b be in the left-hand side union. Then $b \in \operatorname{class} a$ for some $a \in H$. That is, $b = gag^{-1}$ for some $g \in G$. However, $a \in H$ and $H \lhd G$, so $b = gag^{-1} \in H$. Thus $\bigcup_{a \in H} \operatorname{class} a = H$ and so H is a union of conjugacy classes.

Suppose H is a union of conjugacy classes. Say $H = \bigcup_{a \in T} \operatorname{class} a$ for some $T \subseteq G$. We will show that $gHg^{-1} \subseteq H$ for all $g \in G$. Take $g \in G$ and $h \in H$. Then $h \in \operatorname{class} a$ for some $a \in T$. Thus $ghg^{-1} \in \operatorname{class} a \subseteq H$. Thus $H \lhd G$.

3. Let H be a subgroup of G and $N_G(H)$ be the normaliser of H. Prove the following:

   (a) $N_G(H)$ is a subgroup of G.

   (b) $H \lhd N_G(H)$.

   (c) If K is a subgroup of G such that $H \lhd K$, then $K \subseteq N_G(H)$.

   (d) $H \lhd G$ if and only if $N_G(H) = G$.

   Then justify the statement "The normaliser of H is the largest subgroup of G that contains H as a normal subgroup".

(a) This follows from the fact that the normaliser of H is the stabiliser of H under the group action of conjugation by G on the subgroups of G. Recall that the stabiliser of an element is a subgroup of G.

(b) Let $g \in N_G(H)$. Because $g \in N_G(H)$, we have $gHg^{-1} = H$. Thus $H \lhd N_G(H)$.

(c) Suppose K is a subgroup of G and $H \lhd K$. Take $k \in K$. Since H is normal in K, we have $kHk^{-1} = H$. Thus $k \in N_G(H)$ and so $K \subseteq N_G(H)$.

(d) Suppose $H \lhd G$. We know that $N_G(H) \subseteq G$, so it remains to show $G \subseteq N_G(H)$. Let $g \in G$. Since H is normal in G, we have $gHg^{-1} = H$. That is, $g \in N_G(H)$ and so $N_G(H) = G$.

Now suppose that $N_G(H) = G$. Then for all $g \in G$, we have $gHg^{-1} = H$. That is, $H \lhd G$.

4. Let G be a group and H a subgroup of G. The **centraliser** of the subgroup H is defined as $C_G(H) = \{g \in G : gh = hg \text{ for all } h \in H\}$. Prove that $C_G(H) \lhd N_G(H)$.

First we show that $C_G(H)$ is a subgroup of $N_G(H)$. Note that if $g \in C_G(H)$, then $ghg^{-1} = h \in H$ for all $h \in H$. Thus $gHg^{-1} = H$ and so $g \in H$ and $C_G(H) \subseteq N_G(H)$. Note that $1h = h1$ so $1 \in C_G(H)$. Take $a, b \in C_G(H)$. Then, for $h \in H$, we have $(ab)h = ahb = h(ab)$ so $ab \in C_G(H)$. Moreover $ah = ha$ implies $ha^{-1} = a^{-1}h$ so $a^{-1} \in C_G(H)$. Thus $C_G(H)$ is a subgroup of $N_G(H)$.

Now take $n \in N_G(H)$ and $c \in C_G(H)$. We need to show that $ncn^{-1} \in C_G(H)$. That is, $ncn^{-1}h = hncn^{-1}$ for all $h \in H$. Or, equivalently,

$$h = (ncn^{-1})h(ncn^{-1})^{-1} = ncn^{-1}hnc^{-1}n^{-1}.$$

Notice that because $H \triangleleft N_G(H)$, we have $n^{-1}hn \in H$. Then $cn^{-1}hn = n^{-1}hnc$ because $c \in C_G(H)$. This means that $cn^{-1}hnc^{-1} = n^{-1}hn$. From this we can conclude that

$$(ncn^{-1})h(ncn^{-1})^{-1} = nc(n^{-1}hn)c^{-1}n^{-1} = nn^{-1}hnn^{-1} = h,$$

as desired. Thus $C_G(H) \triangleleft N_G(H)$.

5. Let G be a group with subgroups H and K. Prove that if H and K are conjugates then $N_G(H)$ and $N_G(K)$ are conjugates.

Since H and K are conjugates, there is some $g \in G$ such that $K = gHg^{-1}$. We need to show that there is some $g_1 \in G$ such that $N_G(K) = g_1 N_G(H) g_1^{-1}$. The most obvious choice would be $g_1 = g$, so let's try that. So, we are attempting to show $N_G(K) = gN_G(H)g^{-1}$. Let's start by showing $N_G(K) \subseteq gN_G(H)g^{-1}$. Take $a \in N_G(K)$, then $aKa^{-1} = K$. We need to show $a \in gN_G(H)g^{-1}$. That is, $a = gbg^{-1}$ for some $b \in N_G(H)$. Equivalently, $g^{-1}ag = b \in N_G(H)$. We have

$$(g^{-1}ag)H(g^{-1}ag)^{-1} = g^{-1}a(gHg^{-1})ag = g^{-1}(aKa^{-1})g = g^{-1}Kg \subseteq H.$$

Thus $g^{-1}ag \in H$ and so $a \in gN_G(H)g^{-1}$. Therefore, $N_G(K) \subseteq gN_G(H)g^{-1}$. (You might want to take a moment to prove that $K = gHg^{-1}$ implies $H = g^{-1}Kg$.)

Similarly, $g^{-1}N_G(H)g \subseteq N_G(K)$. Thus $N_G(H) = g(g^{-1}N_G(H)g)g^{-1} \subseteq gN_G(K)g^{-1}$. Therefore, $N_G(K) = gN_G(H)g^{-1}$ and so $N_G(H)$ and $N_G(K)$ are conjugates.

Extra problems

(I) Let G be a finite group with exactly two conjugacy classes. Show that $|G| = 2$.

Suppose G is finite with exactly two conjugacy classes. One of the conjugacy classes is $\text{class}\,1 = \{1\}$. Since there is another conjugacy class, there must be some non-identity element $a \in G$ such that $\text{class}\,a \neq \text{class}\,1$. To show that $|G| = 2$, we will show that $|\text{class}\,a| = 1$. Suppose that this is not true. Then $\text{class}\,a$ is not a singleton conjugacy class and so $a \notin Z(G)$. Thus $Z(G) = \{1\}$. From the Class Equation, we have

$$|G| = 1 + |\text{class}\,a| = 1 + |G : C_G(a)|.$$

By Lagrange's Theorem (G is finite, so we can apply it), we have $|G : C_G(a)| = |G|/|C_G(a)|$. That is, $|G| = |G : C_G(a)||C_G(a)|$. Therefore,

$$|G : C_G(a)||C_G(a)| = 1 + |G : C_G(a)|.$$

Then $|G : C_G(a)|$ divides 1 (since it divides all other terms). This means that $|\text{class}\,a| = |G : C_G(a)| = 1$, contradicting our assumption. Therefore, we must have that $\text{class}\,a$ is a singleton and so $|G| = 2$.

(II) Prove that if G is a finite group and $a \in G$ such that $|\text{class}\,a| = 2$, then G is *not* simple.

Note: Recall that a group is called simple if its only normal subgroups are itself and the trivial subgroup.

Recall that $\text{class}\,a$ is the orbit of a. By the Orbit-Stabiliser Theorem, $|\text{class}\,a| = |G * a| = |G : S(a)|$. Thus $S(a)$ is a subgroup of G of index 2. So $S(a)$ is a normal subgroup of G (Theorem 1.34). We will now try to show that $S(a) \neq \{1\}$ and $S(a) \neq G$. If $S(a) = \{1\}$, then $|G : S(a)| = |G|$ and so $|G| = 2$. All groups of order 2 are isomorphic to $\mathbb{Z}_2$, an abelian group. However, if G is abelian, then all conjugacy classes are singletons but $|\text{class}\,a| = 2$. If $S(a) = G$, then $|G : S(a)| = 1$, a contradiction because $|G : S(a)| = |\text{class}\,a| = 2$. Thus, $S(a)$ is a normal subgroup of G such that $S(a) \neq G$ and $S(a) \neq \{1\}$. That is, G is not simple.